

Get KIVA Running & Start Playing

No technical experience needed

MA AI Hub x SundAI Pre-Session

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Before You Start – Check These

Mac users

- Any Mac with Apple Silicon (M1/M2/M3) works great
- Open **Terminal** app (Cmd+Space, type “Terminal”)

Linux users

- Ubuntu 20.04+ recommended
- Open your terminal app

Windows is not supported

If you only have Windows, use the **GitHub Codespace** option (we'll help you set it up).

You will need:

- An **OpenAI API key** – get one at <https://platform.openai.com/api-keys>
- **Google Chrome** browser (version 134 or newer)

Step 1 of 6: Download KIVA

Open your terminal and paste this line, then press Enter:

```
git clone https://github.com/sensein/riverst.git
```

Then type:

```
cd riverst
```

What just happened?

You downloaded the entire KIVA project to your computer and moved into its folder.

Step 2 of 6: Create a Python Environment

Paste these **3 lines one at a time**, pressing Enter after each:

❶ `cd src/server`

❷ `conda create -n riverst python=3.11 -y`

(This downloads Python. It takes 1–2 minutes. Wait for it to finish.)

❸ `conda activate riverst`

(You should see “(riverst)” appear at the left of your terminal prompt.)

Don't have conda?

Install it first: <https://docs.conda.io/en/latest/miniconda.html>

Download Miniconda, install it, then come back here.

Step 3 of 6: Install Server Packages

Paste these **2 lines one at a time**, pressing Enter after each:

❶ `conda install -c conda-forge ffmpeg -y`

(Audio processing tool. Takes 1–2 minutes.)

❷ `pip install -r requirements.txt`

(All Python dependencies. Takes 2–5 minutes.)

Wait for each command to finish

You'll see a lot of text scrolling by. That's normal. Wait until the prompt returns before pasting the next line.

Step 4 of 6: Add Your API Key

Paste this line and press Enter:

```
cp env.example .env
```

Now open the file `.env` in any text editor (VS Code, nano, TextEdit):

```
nano .env
```

Find the line that says `OPENAI_API_KEY=""` and paste your key between the quotes:

```
OPENAI_API_KEY="sk-proj-abc123..."
```

Save and close (in nano: Ctrl+X, then Y, then Enter).

Step 5 of 6: Install the Frontend

Open a **new, second terminal window** and paste these lines one at a time:

❶ `cd riverst/src/client/react`

❷ `npm install`

(Downloads web app dependencies. Takes 1–3 minutes.)

Don't have npm?

Install Node.js first: <https://nodejs.org/> (download the LTS version)

Step 6 of 6: Start KIVA!

You need **two terminal windows** running at the same time.

Terminal 1 (Server)

```
cd riverst/src/server  
conda activate riverst  
python main.py
```

Wait for "Uvicorn running on..."

Terminal 2 (Client)

```
cd riverst/src/client/react  
npm run dev
```

Wait for "Local: http://localhost:5173"

Start the server FIRST!

Terminal 1 must show "Uvicorn running" before you run Terminal 2.

Open Chrome and go to: `http://localhost:5173`

What Can You Do With KIVA?

Once KIVA is running, the homepage shows you **4 categories of activities**:

- 1  **Avatar customization** – Pick or create your avatar
- 2  **Playground** – Free chat or a quick demo
- 3  **KIVA activities** – Audiobooks & vocabulary training
- 4  **L2 learning** – English or Italian as a second language

Start here

Click **“Select your avatar”** first, then try **“Demo ready to go”**.

Quick Start: Avatar + Demo in 5 Clicks

- 1 Click **“Select your avatar”**
- 2 Click on **Fabio** or **Bruce** to choose one
- 3 Click **Back** to return to the homepage
- 4 Under “Playground”, click **“Free style interaction”**
- 5 Click **“Run a demo”** → allow microphone → start talking!

Tip

Use **Chrome** and make sure your microphone is enabled. The avatar listens and responds in real time.

Create Your Own Custom Session

The “**Free style interaction**” activity lets you customize your avatar’s behavior.

When you click it, you see a form with these key fields:

Avatar System Prompt **Most important!** This tells the avatar WHO to be.

Example: “You are a friendly math tutor for 8-year-olds.”

Task Description What the avatar should DO during the session.

Example: “Help the student practice multiplication tables.”

Avatar Personality The avatar’s character traits.

Example: “Patient, encouraging, uses fun examples.”

Camera Settings How close the camera is: head / upper / mid / full

Embodiment Avatar (3D character) or Waveform (audio only)

Custom Session: Step by Step

- 1 On the homepage, click **“Free style interaction”**
- 2 In the form, find **“Avatar System Prompt”**
- 3 Replace the default text with your own, e.g.:
`You are a friendly science tutor. Keep responses short.`
- 4 Optionally change **“Task Description”** and **“Avatar Personality”**
- 5 Click **“Run a demo”**
- 6 Allow microphone access and start talking!

That's it!

You just created a custom AI tutor session. The avatar follows the persona you defined.

Example Custom Sessions You Can Try

Idea	System Prompt	Task Description
Math Tutor	You are a patient math tutor for kids. Use simple language.	Practice addition and subtraction with the student.
Storyteller	You are a creative storyteller. Make up stories from what the user says.	Tell an interactive story where the user chooses what happens.
Language Coach	You are a Spanish teacher. Speak slowly and repeat words.	Teach 5 basic Spanish greetings and practice them.
Career Advisor	You are a career counselor. Ask about interests and suggest careers.	Help the user explore career paths based on their passions.

KIVA Activities: Vocabulary Training

KIVA includes **structured tutoring activities** with built-in lesson flows:

- **Child vocabulary training** – Guided play to discover new words
- **English for Italians** – ESL vocabulary with avatar support
- **Italian for Americans** – ISL vocabulary with avatar support
- **Audiobook** – Listen and follow along with visual cues

How is this different from free style?

These activities have a **structured flow**: the avatar follows a lesson plan with stages, tracks progress, and adapts to the learner. Free style is open-ended conversation.

- **“Failed to create session”** – Check OpenAI key in `src/server/.env`
- **Avatar not loading** – Clear browser cache or try DevTools console
- **No microphone** – Chrome Settings → Privacy → Microphone → Allow
- **“conda: command not found”** – Install Miniconda first (see Step 2)
- **“npm: command not found”** – Install Node.js first (see Step 5)
- **Port in use** – Restart your computer, or kill processes on 7860/5173
- **ffmpeg error** – Run `conda install -c conda-forge ffmpeg`

Quick Reference: All Commands

1. `git clone https://github.com/sensein/riverst.git && cd riverst`

2. `cd src/server && conda create -n riverst python=3.11 -y && conda activate riverst`

3. `conda install -c conda-forge ffmpeg -y && pip install -r requirements.txt`

4. `cp env.example .env` then add your OpenAI key to `.env`

5. `cd ../client/react && npm install`

6. Terminal 1: `python main.py` | Terminal 2: `npm run dev`

You're ready!

Install → Pick avatar → Customize → Talk

 <https://github.com/sensein/riverst>

 Ask for help on Slack anytime